

No.	Name	Description	Priority	Rationale	Author
User					
1		The user shall create an account		Allows us to keep track of user	Jose Felten
1.1		- The user shall provide their username		Allows user to have a username	Jose Felten
1.2		- The user shall provide a secure password		Ensures security	Jose Felten
1.3		- The user shall confirm their password		Allows user to ensure password is what they want it to be	Jose Felten
1.4		- The user shall provide their phone number		Allows for MFA	Jose Felten
1.5		- The user shall verify their phone number		Ensures MFA is secure	Jose Felten
2		The system shall use the device location to find nearby relief systems.		Nearby centers help users quickly locate available help.	Kevin Beni
2.1		- The system shall alert users about hurricanes near saved locations		A short message gives the users clear information about the nearby hurricane.	Kevin Beni
2.2		- The system shall show each relief center's operating hours		Open status information prevents users from traveling to a closed relief center.	Kevin Beni
12		The user shall set notification preferences for hurricane alerts.		1 Users need control over how they receive time-sensitive alerts so they don't miss criti	Jon Liu
12.1		- The user shall choose to receive alerts via push notification, SMS, or email		2 Offering multiple channels ensures the user gets alerts even if one method is unavaila	Jon Liu
13		The user shall add emergency contacts to their profile.		2 Emergency contacts allow loved ones to be notified automatically during a crisis.	Jon Liu
Admin					
3		The user shall moderate misuse of relief requests		Ensures security and integrity of the app	Jose Felten
3.1		- The user shall delete misleading relief requests		Ensures integrity	Jose Felten
3.2		- The user shall block problematic users		Ensures our app is being used for valid reasons	Jose Felten
14		The system shall provide admins with a dashboard showing active relief requests		2 A dashboard lets admins quickly assess how well the relief effort is functioning and w	Jon Liu
15		The admin shall be able to add, edit, or remove relief center listings.		1 Relief center information changes quickly during a disaster, so admins need direct co	Jon Liu
16		The system shall allow admins to generate a summary report of relief requests at		3 Reports help admins and stakeholders evaluate the relief effort after the fact.	Jon Liu
16.1		- The admin shall review new relief request before they are posted		2 Review helps keep incorrect requests from being shared, especially from bots.	Kevin Beni
		- The admin shall mark a relief request as resolved when help is completed.		1 Resolved requests are removed from from the active work list.	Kevin Beni
Volunteer					
4		The user shall enter their full name		Allows app to have user full name	Jose Felten
5		The user shall enter their home address		Allows us to map out users	Jose Felten
6		The user shall enter relevant skills		Allows us to best provide to user a good way help out	Jose Felten
7		The user shall view relief requests corresponding to relevant skills		Allows users to best use their skillset	Jose Felten
8		The user shall respond to relief requests		Allows user of application to help	Jose Felten
17		The user shall set their availability schedule for volunteering.		2 Knowing when a volunteer is available helps match them to relief requests they can a	Jon Liu
17.1		- The user shall be able to mark availability as recurring or one-time.		3 Recurring availability saves volunteers from re-entering the same schedule repeatedl	Jon Liu
18		The user shall view a history of the relief requests they have responded to, along		2 A visible history gives volunteers a sense of their impact and accountability for their v	Jon Liu
		The user shall filter relief requests by distance		2 Distance filters help volunteers prepare before helping others.	Kevin Beni
		The user shall report when they are unable to complete a request.		1 The request can quickly be made available to another user.	Kevin Beni
Request					
9		The user shall submit relief requests		Allows users to provide things that need to be fixed	Max Vogel
9.1		- The user's geographical location shall be automatically obtained		Allows users to view people in need near to them	Max Vogel
9.2		- The user shall enter their location manually if location services unavailable		Allows user to still find people in need near them if services go down.	Max Vogel
9.3					Max Vogel
19		The user shall be able to view the current status of a submitted relief request (e.g.		1 Status visibility reduces anxiety and repeated inquiries from people waiting for help.	Jon Liu
19.1		- The system shall notify the requester when their request's status changes.		2 Automatic notifications keep the requester informed without them needing to check m	Jon Liu
20		The user shall be able to cancel a submitted relief request.		2 Canceling prevents volunteers from wasting time responding to a need that no longer	Jon Liu
21		The user shall be able to attach photos to a relief request to better illustrate their		3 Photos help volunteers and admins understand the severity and nature of a request. #	Jon Liu
		The user shall enter the number of people who need help.		3 The number helps first responders prepare enough supplies.	Kevin Beni
		The system shall show a sample list of information needed before a request is		2 The list helps users submit complete requests.	Kevin Beni
Hurricane Tracking					
10		The user shall view the location of a hurricane		Allows users to see the hurricane	Jose Felten
11		The location of a hurricane shall update periodically		Allows users to see an update of where the hurricane is at	Jose Felten
22		The system shall display recommended evacuation routes for a user's area durin		1 Clear evacuation guidance can directly save lives during a storm.	Jon Liu
22.1		- The system shall update the recommended route if road closures are reported.		2 Static routes could lead users into unsafe or blocked roads.	Jon Liu
		The system shall display the hurricane's direction of travel		2 A checklist that helps users prepare before a hurricane.	Kevin Beni
		The system shall show the severity level of the hurricane.		2 Travel direction helps users understand where the storm is moving.	Kevin Beni
				3 The storm information alerts users to take quicker action.	Kevin Beni
23		The user shall be able to view historical hurricane data for their saved locations.		3 Historical data helps users understand their area's risk level and prepare accordingly.	Jon Liu
Emergency Communication					
24		The user shall be able to trigger an SOS alert that notifies nearby volunteers and		1 An SOS feature provides a fast path to help when a situation is too urgent to wait for	Jon Liu
24.1		- The system shall share the user's live location with their emergency contacts for		2 Sharing location with trusted contacts adds a layer of safety without permanently exp	Jon Liu
25		The system shall provide critical hurricane safety information in multiple language		2 Disaster-prone communities are often linguistically diverse, and safety information mu	Jon Liu
25.1		- The system shall support, at minimum, English and Spanish.		3	Jon Liu
		The user shall choose the best way for a volunteer to arrive		3 A preferred time helps avoid missed visits.	Kevin Beni
		The system shall let the user choose which emergency contacts receive a		3 The user can notify the most appropriate contacts.	Kevin Beni
		The system displays local emergency contacts for users.		3 Phone numbers provide a direct way to reach emergency help.	Kevin Beni
Non-Functional					
26		The system shall use high-contrast color schemes to remain readable in bright		This allows for users to read the text in situations where the lighting might be bad	Max Vogel
27		The UI shall use universally recognizable icons (SOS, hurricane, medical) to		The UI simplifies things for the user	Max Vogel
27		reduce reliance on text.		Users who use this app could have one of their hands injured while using.	Max Vogel
28		The system shall be operable one-handed		Allows max availability of use for the app.	Max Vogel
29		The system shall maintain 99.9% uptime during active hurricane events			Max Vogel
30		(Category 1+)		Protects user information and ensures confidentiality and integrity.	Max Vogel
31		All user location data shall be encrypted in transit and at rest.		Allows for many different users at once	Max Vogel
32		The system shall support at least 10,000 concurrent users during a major storm		Allows for user to use the app when they are unable to type	Max Vogel
32		event without degraded response time.			Max Vogel
33		The system shall support voice input for submitting relief requests		Ensures the availability of the app	Max Vogel
34		The system shall auto-scale backend infrastructure in response to traffic spikes		Allows for open source DB and allows for Post GIS.	Max Vogel
34		without manual intervention.		I have been wanting to learn AWS for a while now, this would be a great way to do	Max Vogel
35		PostgreSQL DB shall be used for this application		Ensures passwords are safe even in case of a data breach.	Max Vogel
36		AWS shall be implemented into the application, AWS Pinpoint, AWS RDS, etc.		Allows for us to respond and fix whatever problem is happening.	Max Vogel
37		User passwords will never be stored persistently, only the hashed values.		We do not want to be spending a crazy amount of money.	Max Vogel
39		The app shall alert the developers if there is any type of error in the application.		Allows us to see if there is anything out the ordinary.	Max Vogel
40		System shall be designed to operate within our budget range.		Users may not have access to charge their devices for prolonged periods of time.	Max Vogel
41		System shall provide dashboard to devs showing health of app			Max Vogel
41		The app shall minimize battery consumption during active hurricane tracking			Max Vogel

